

Course milestone M5: Pathway Declaration and Mandated Contrast

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m05.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M05**. Brightspace carries the deadline.

This milestone presents **Book Milestone 5: Your pathway, declared** (checkpoint, version 1) of the course book, EDR|AI.

Open the milestone workbook in Colab

This chapter closes Studio 5: Develop the pathway. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. Research Contract v1: objective by target and reach by data strategy by warrant, with the pathway declared and its limits written before any result exists.

What you bring

You also carry forward the last milestone's artifact: **Milestone 4: Your research contract, v0**. This one builds on it, and the chain assumes it is versioned and filed.

These pieces are conditional. Bring each one when its condition is yours; when it is not, the lesson's own drill or a plain skip is the right call, and the milestone does not require the piece.

- Lesson 14: Observational Descriptive Research: either your four boxes and selection diagram, if this is your pathway, or the classification drill that rules it out.
- Lesson 15: Observational Causal Research: either your causal diagram with its leverage named and priced, or the drill that shows why this road is not yours.
- Lesson 16: Experimental Descriptive Research: either your revealed-characteristic estimand with its demand and instrument threats named, or the drill run on your own question.
- Lesson 17: Prediction and Generalization: either your signed four-part forecasting contract with its boundary, or the contrast drill: the plain rejection line, the prediction-time leakage check run on your own features, and two sentences on why prediction answers a different question.
- Lesson 18: Experimental Causal Research: either your treatment, outcome, and potential outcomes with the nearest threat named, or the drill plus the confounder that blocks the causal reading.
- Lesson 19: Hybrid and Complex Designs: if your design has stages: the stage-by-stage map, its weakest link, and the one-sentence justification for keeping or cutting it.

One bound on the skip rule here: this milestone requires your PRIMARY pathway's full piece plus ONE contrast drill from any other route you choose: reading a road that is not yours is what makes yours visible; skipping the rest is correct. The overlay binds only when your design has stages.

The practice

1. Use the route guide above to choose your pathway, then read that lesson and at least one that does not fit, so you can say why yours does.
2. Declare your pathway and write the crossing that licenses it: what makes your comparison mean what you claim.
3. Write the limits: what this pathway cannot establish no matter how clean the execution.
4. Update your claim boundary to match the pathway, and mark the change as a new Contract version.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** A change of pathway can change your permission status; recheck rather than assume.
- **Evidence, provenance, and reproducibility.** Name the prior work that used this pathway on a comparable question.
- **AI activity, verification, and human decisions.** Ask an assistant to argue for a different pathway, then answer the argument yourself.
- **Uncertainty, claim boundary, and revision history.** Different pathways carry different sources of uncertainty; name yours here.

Where this milestone sits

You arrived carrying Milestone 4: Your research contract, v0. Milestone 6 puts data and measurement under governance so the declared pathway has something trustworthy to run on. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Realized data that will not support the declared pathway sends you back here before analysis, not after.

How this milestone is assessed

Each row scores **0, 1, or 2. 10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: Research Contract v1: objective by target and reach by data strategy by warrant, with the pathway declared and its limits written before any result exists	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

! Blocking gate

No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.